31. Bearings

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Three-figure bearings:

Write the bearing for a direction that is 42° measured clockwise from North, as a three-figure bearing.

Answer: A bearing is always written using three digits, so 42° is written as **042°**.

Example 2 — Compass to bearing:

What is the approximate three-figure bearing for South-West (SW)?

Answer: SW lies exactly halfway between South (180°) and West (270°), so its bearing is **225°**.

Example 3 — Back bearings:

A hiker walks from point A to point B on a bearing of 118°. Find the bearing of A from B (the back bearing).

Answer: Since 118° is less than 180°, add 180°: 118° + 180° = **298°**. The back bearing of A from B is 298°.

Example 4 — Compass direction from a bearing:

A bearing of 118° is roughly closest to which of the 8 main compass points?

Answer: 118° is closest to South-East (135°), so the direction is roughly **South-East**.

Example 5 — Multi-step: turning on a bearing:

A cyclist rides on a bearing of 110°, then turns 90° clockwise. What bearing is she now cycling on?

Answer: Turning clockwise adds to the bearing: 110° + 90° = **200°**.

Foundation

15 questions

1. What is the bearing of North? Give your answer as a three-figure bearing.

2. What is the bearing of East? Give your answer as a three-figure bearing.

3. Write 6° as a three-figure bearing.

4. Write 260° as a three-figure bearing. A) 26 B) 026 C) 260 D) 0260

5. A ship sails on a bearing of 090°. In which compass direction is it travelling?

6. What is the bearing of South, as a three-figure bearing?

7. A plane flies on a bearing of 180°. In which compass direction is it travelling? A) North B) East C) South D) West E) North-East

8. List the 8 compass directions in clockwise order starting from North.

9. What is the approximate bearing of South-East (SE)?

10. A walker is told to head on a bearing of 225°. Which compass direction does this match? A) North-East B) South-East C) South-West D) North-West E) West

11. Select ALL that apply: which of the following are valid three-figure bearings? A) 060° B) 9° C) 210° D) 088° E) 380°

12. What is the bearing of West, as a three-figure bearing?

13. A hiker walks on a bearing of 090° and then turns to walk on a bearing of 270°. What is the relationship between these two directions?

14. Select ALL that apply: which of these bearings point in a generally southerly direction (between 090° and 270°)? A) 050° B) 160° C) 190° D) 290° E) 340°

15. A compass needle points on a bearing of 000°. Which compass direction is this? A) North B) East C) South D) West E) North-East

Standard

20 questions

1. A ship's radar shows a lighthouse at a bearing of 040° from the ship. Describe, using a compass direction, roughly where the lighthouse lies relative to the ship.

2. The bearing of town B from town A is 055°. What is the back bearing (the bearing of A from B)?

3. The bearing of a hill from a campsite is 132°. What is the back bearing of the campsite from the hill?

4. A bearing of 290° is given. What is the back bearing? A) 090° B) 100° C) 110° D) 120° E) 130°

5. A boat sails from harbour P to harbour Q on a bearing of 200°. What is the bearing from Q back to P?

6. Select ALL that apply: which of these pairs correctly show a bearing and its exact back bearing? A) 030° and 210° B) 085° and 175° C) 150° and 330° D) 240° and 060° E) 290° and 100°

7. A bearing of 065° is measured from North, turning clockwise. Through how many degrees would you turn, starting by facing North, to face along this bearing?

8. Two signposts show that Hilltown is on a bearing of 118° from Riverside. What direction, roughly (to the nearest compass point of N, NE, E, SE, S, SW, W, NW), is Hilltown from Riverside? A) East B) South-East C) South D) South-West E) West

9. A bearing of 200° is given. What is the back bearing?

10. A plane flies on a bearing of 052° from airport X to airport Y. Describe the back bearing in words (using a compass direction) and give its value as a three-figure bearing.

11. A bearing of 004° is given. What is the back bearing? A) 094° B) 184° C) 186° D) 188° E) 274°

12. Select ALL that apply: a ship travels from a buoy to a jetty on a bearing of 085°. Which statements about the return journey (jetty to buoy) are true? A) The back bearing is 265°. B) The back bearing is found by subtracting 180° from 085°. C) The back bearing is found by adding 180° to 085°. D) The return journey is on a bearing of less than 180°. E) 085° and 265° differ by exactly 180°.

13. A bearing from a ship to a buoy is measured as 218°. What is the bearing from the buoy back to the ship?

14. On a map, a church is due North-West of a school. Give the bearing of the church from the school.

15. A diagram shows two points, C and D. The bearing of D from C is 325°. State the bearing of C from D.

16. The bearing of town C from town D is 095°. What is the back bearing (the bearing of D from C)?

17. The bearing of a bridge from a village is 160°. To the nearest compass point, roughly what direction is the bridge from the village?

18. A bearing of 275° is given. What is the back bearing? A) 095° B) 085° C) 105° D) 115°

19. On a map, a park is due South-East of a house. Give the bearing of the park from the house.

20. The bearing of point E from point F is 168°. State the bearing of F from E.

Hard

5 questions

1. A hiker walks from camp to a stream on a bearing of 050°. What bearing must she walk on to get back from the stream to the camp?

2. From one starting point, a drone flies off on a bearing of 060° while a second drone flies on a bearing of 170°. How big is the angle between their flight paths?

3. A cyclist rides on a bearing of 110°, then turns 90° clockwise. What bearing is she now cycling on?

4. Select ALL that apply: the bearing from a house to a park is 145°. Which of these statements are true? A) The back bearing (park to house) is 325°. B) The park lies roughly to the south-east of the house. C) The back bearing is found by subtracting 180° from 145°. D) The house lies roughly to the north-west of the park. E) The angle between the bearing and its back bearing is 180°.

5. Two ships leave the same harbour at the same time: Ship 1 on a bearing of 030°, Ship 2 on a bearing of 300°. What is the angle between their two courses?

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